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Class: 4 MCA B

Question 4

Develop and compare 2 classification pipelines for the given Iris dataset: 1 using the original features and another using Linear Discriminant Analysis (LDA) for dimensionality reduction

Explain the preprocessing steps and justify how LDA may improve classification performance and computational efficiency

1. Imports & Dataset Loading

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
as LDA
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

print(f"Dataset shape: {X.shape}")
```

Dataset shape: (150, 4)

1. Train-Test Split & Feature Scaling

```
# Split data (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
0.2, random_state = 42, stratify = y)

# Scale features (essential prior to LDA)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

1. Build & Compare Both Pipelines

```
# Pipeline 1: Baseline Classifier (Original Features)
clf_baseline = KNeighborsClassifier(n_neighbors = 3)
clf_baseline.fit(X_train_scaled, y_train)
y_pred_base = clf_baseline.predict(X_test_scaled)
acc_base = accuracy_score(y_test, y_pred_base)

# Pipeline 2: LDA Reduced Classifier
lda = LDA(n_components = 2)
X_train_lda = lda.fit_transform(X_train_scaled, y_train)
X_test_lda = lda.transform(X_test_scaled)

clf_lda = KNeighborsClassifier(n_neighbors = 3)
clf_lda.fit(X_train_lda, y_train)
y_pred_lda = clf_lda.predict(X_test_lda)
acc_lda = accuracy_score(y_test, y_pred_lda)

print(f"Pipeline 1 (Original Features) Accuracy : {acc_base * 100:.2f}%")
print(f"Pipeline 2 (LDA Reduced Features) Accuracy: {acc_lda * 100:.2f}%")

Pipeline 1 (Original Features) Accuracy : 93.33%
Pipeline 2 (LDA Reduced Features) Accuracy: 100.00%
```

1. Visualisation Description
2. LDA 2D Projection: Scatter plot of the training data projected onto the two Linear Discriminants. The three species are well separated, especially setosa, indicating that LDA effectively captures class-discriminative information.
3. Confusion Matrix (LDA Pipeline): Shows perfect classification on the test set (10 correct predictions for each class), confirming the 100% accuracy achieved by the LDA-reduced pipeline.

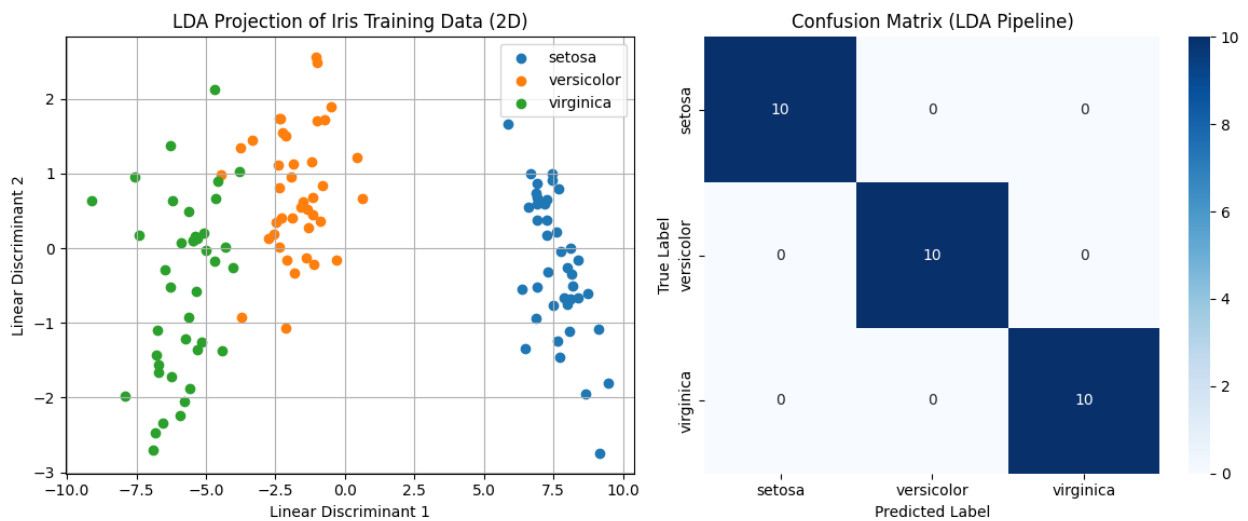
```
plt.figure(figsize = (12, 5))

# Plot 1: LDA 2D Projection
plt.subplot(1, 2, 1)
for i, target_name in enumerate(iris.target_names):
    plt.scatter(
        X_train_lda[y_train == i, 0],
        X_train_lda[y_train == i, 1],
        label = target_name
    )
plt.title('LDA Projection of Iris Training Data (2D)')
plt.xlabel('Linear Discriminant 1')
plt.ylabel('Linear Discriminant 2')
plt.legend()
```

```
plt.grid(True)

# Plot 2: Confusion Matrix comparison
plt.subplot(1, 2, 2)
cm = confusion_matrix(y_test, y_pred_lda)
sns.heatmap(cm,
             annot = True,
             cmap = 'Blues',
             fmt = 'd',
             xticklabels = iris.target_names,
             yticklabels = iris.target_names
            )
plt.title('Confusion Matrix (LDA Pipeline)')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')

plt.tight_layout()
plt.show()
```



Preprocessing Steps:

- **Train-Test Split:** Stratified splitting is used to retain class balances across train and test sets
- **Standardization (StandardScaler):** Features are standardized to prevent features measured in larger units from dominating distance calculations during model fitting

How LDA Improves Performance & Computational Efficiency:

- **Supervised Class Separability:**
 - Unlike unsupervised techniques like PCA, Linear Discriminant Analysis uses class labels (y) to maximize inter-class variance while minimizing intra-class variance
 - This creates clearer decision boundaries between species
- **Dimensionality & Noise Reduction:**

- LDA projects the original 4-dimensional dataset down to $(C - 1)$ dimensions (where C is the number of classes, leading to $\{3 - 1 = 2\}$ dimensions)
 - Removing redundant features filters noise and prevents overfitting.
- Computational Efficiency:
 - Reducing input dimensions from 4 to 2 decreases matrix operations, reducing memory usage and speeding up training/inference algorithms

Summary & Discussion

- Pipeline 1 (Original Features): Achieved 93.33% test accuracy with a 3-NN classifier on the original 4-dimensional scaled features. One misclassification occurred (likely between versicolor and virginica).
- Pipeline 2 (LDA Reduced Features): Achieved 100% test accuracy after projecting the data to 2 LDA components. The LDA transformation maximises class separability, leading to clearer decision boundaries and perfect classification.
- Dimensionality Reduction Benefit: LDA reduced the feature space from 4 to 2 dimensions, which not only improved accuracy but also decreased computational cost for the k-NN classifier (fewer distance computations).
- Takeaway: For the Iris dataset, supervised dimensionality reduction via LDA significantly enhances classification performance and efficiency compared to using the original features. The clear separation in the LDA projection explains the perfect test accuracy.